

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Security	USN-1	As an authorized grid analyst, I can log into the dashboard securely using specialized credentials..	2	High	Developer 1
Sprint-1	Data Ingestion	USN-2	As an analyst, I can upload massive raw CSV consumption files (16,800+ rows) into the data pipeline.	5	High	Developer 2
Sprint-1	Data Cleaning	USN-3	As an analyst, I want a Python script to automatically clean null timestamps and isolate outlier values.	5	High	Developer 1

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Regional Separation	USN-4	As an analyst, I want the dataset structured into 5 distinct national zones (North, West, South, East, North-East).	3	Medium	Developer 2
Sprint-2	Visual Interface	USN-5	As a director, I want an interactive map interface in Tableau to dynamically select and view each of the 5 zones.	5	High	Developer 1
Sprint-2	Query Optimization	USN-6	As an analyst, I want Context Filters implemented so that state-level sub-rankings remain mathematically accurate.	5	High	Developer 2
Sprint-2	Timeline Syncing	USN-7	As a policy maker, I want a dual-year timeline view to cross-examine 2019 baseline trends with 2020 lockdown drops.	5	Medium	Developer 1
Sprint-2	Executive Export	USN-8	As a director, I can export custom-filtered dashboard charts directly as PDF summary reports for meetings.	3	Low	Developer 2

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	15	6 Days	1 July 2026	06 Jul 2026	15	06 Jul 2026
Sprint-2	18	6 Days	08 Jul 2025	13 Jul 2026	18	13 Jul 2026

Velocity:

Scenario Context: Team velocity is **20 story points** per **10-day sprint**.

Average Velocity (AV) Formula:

$$\text{Average Velocity(AV)} = \frac{\text{Total Story Points Per Sprint}}{\text{Sprint Duration in Days}}$$

 **Calculation:**

$$AV = \frac{20 \text{ Points}}{10 \text{ Days}} = 2 \text{ story points per day}$$

Burndown Chart Analysis

Chart Definition: The Burndown Chart is a visual representation of work remaining versus time. It tracks the progress of the team by displaying the reduction of Story Points as tasks move from "In Progress" to "Done" throughout the sprint duration.

Performance Interpretation:

- **X-Axis (Time):** Represents the 6-day duration of the sprint, tracking work progression day-by-day.
- **Y-Axis (Work Remaining):** Represents the total volume of Story Points remaining (starting at 15 points for Sprint-1 and 18 points for Sprint-2).
- **Interpretation:** The **Ideal Progress Line** tracks downwards diagonally toward zero. In our project, significant drops in the graph correlate with the completion of the core Python data-processing engine in Sprint-1 and the successful implementation of Context Filters in Sprint-2. This demonstrates a steady, high-velocity workflow, reducing remaining work to zero by the planned sprint end date.